

$$A = \{9, 5, 6, 8, 10\} \quad B = \{1, 2, 3, 7, 9\} \quad C = \{1, 0\}$$

Prove:

$$\textcircled{1} A \cup (B \cap C) = \{0, 1, 2, 3, 5, 6, 7, 8, 9, 10\} = (A \cup B) \cup C$$

$$\textcircled{2} A \cap (B \cap C) = \emptyset = (A \cap B) \cap C$$

$$\textcircled{3} A \cup (B \cap C) = \{1, 5, 6, 8, 9, 10\} = (A \cup B) \cap (A \cup C)$$

$$\textcircled{4} A \cap (B \cup C) = \{9\} = (A \cap B) \cup (A \cap C)$$

